

Remote Perimeter Mapping

Identify and Inventory Remote Access Devices, Remote User Endpoints, and Entry Points into the Network

Introduction

Remote perimeter mapping is the process of identifying and documenting all remote access entry points within a network. These entry points include Secure Shell (SSH), Remote Desktop Protocol (RDP), VPN gateways, HTTPS services, and other remote management interfaces. Identifying these services helps security professionals understand the network attack surface, monitor exposed services, and improve the overall security posture of an organization.

Aim

To identify and inventory remote access devices, remote user endpoints, and entry points into the network using Nmap.

Objectives

- Install and verify the Nmap scanning tool.
- Verify network connectivity between the scanning and target systems.
- Scan common remote access ports.
- Identify services running on non-default ports.
- Classify remote access devices and entry points.

System Requirements

- Kali Linux Virtual Machine (Scanner)
- Ubuntu Linux Virtual Machine (Target)
- VMware Workstation
- Nmap
- Both VMs connected to the same virtual network

Network Configuration

Machine	Role	IP Address
Kali Linux	Scanner	192.168.80.128
Ubuntu Linux	Target Server	192.168.80.129

Expected Output

- Successful installation of Nmap.
- Verification of network connectivity.
- Identification of common remote access ports.
- Discovery of services running on non-standard ports.
- Documentation of remote access entry points.

Procedure

Step 1: Install Nmap

Command:

```
sudo apt update  
sudo apt install nmap -y
```

Explanation: The package list is updated, and the Nmap utility is installed on the Kali Linux machine. Nmap is used for network discovery and security auditing by scanning hosts, ports, and services.

```
(kali㉿kali)-[~]  
$ sudo apt update  
sudo apt install nmap -y  
[sudo] password for kali:  
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]  
Get:4 http://kali.download/kali kali-rolling/non-free amd64 Packages [169 kB]  
Fetched 74.8 MB in 14s (5,250 kB/s)  
28 packages can be upgraded. Run 'apt list --upgradable' to see them.  
nmap is already the newest version (7.99+dfsg-1kali1).  
nmap set to manually installed.  
The following packages were automatically installed and are no longer required:  
  firmware-marvell-prestera libntfs-3g89t64 libopenjph0.27 libvidstab1.1 libx265-215  
Use 'sudo apt autoremove' to remove them.
```

Step 2: Verify Nmap Installation

Command:

```
nmap --version
```

Explanation: This command verifies that Nmap has been installed successfully and displays the installed version.

```
(kali㉿kali)-[~]  
$ nmap --version  
Nmap version 7.99 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.8 openssl-3.6.3 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6  
Compiled without:  
Available nsock engines: epoll poll select
```

Step 3: Verify Ubuntu IP Address

Command:

```
hostname -I
```

Explanation: Displays the IP address of the Ubuntu target machine.

```
eswar@ubuntu:~$ hostname -I  
192.168.80.129
```

Step 4: Verify Kali Linux IP Address

Command:

```
hostname -I
```

Explanation: Displays the IP address of the Kali Linux scanning machine.

```
(kali㉿kali)-[~]  
$ hostname -I  
192.168.80.128
```

Step 5: Verify Network Connectivity

Command:

```
ping 192.168.80.129
```

Explanation: Checks whether the Ubuntu machine is reachable from Kali before performing network scanning.

```
(kali㉿kali)-[~]
$ ping 192.168.80.129
PING 192.168.80.129 (192.168.80.129) 56(84) bytes of data.
64 bytes from 192.168.80.129: icmp_seq=1 ttl=64 time=2.20 ms
64 bytes from 192.168.80.129: icmp_seq=2 ttl=64 time=1.38 ms
64 bytes from 192.168.80.129: icmp_seq=3 ttl=64 time=1.25 ms
64 bytes from 192.168.80.129: icmp_seq=4 ttl=64 time=0.792 ms
^C
— 192.168.80.129 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 0.792/1.405/2.200/0.508 ms
```

Step 6: Scan Common Remote Access Ports

Command:

```
nmap -p 22,3389,443 192.168.80.129
```

Explanation: This command scans the commonly used remote access ports:

- Port 22 – SSH
- Port 443 – HTTPS
- Port 3389 – Remote Desktop Protocol (RDP)

The scan determines whether these services are available on the target system.

Observed Result:

Port	Service	Status
22	SSH	Closed
443	HTTPS	Closed
3389	RDP	Closed

```
(kali㉿kali)-[~]
$ nmap -p 22,3389,443 192.168.80.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-30 01:09 -0400
Nmap scan report for 192.168.80.129
Host is up (0.0022s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
443/tcp    closed https
3389/tcp   closed ms-wbt-server
MAC Address: 00:0C:29:96:C2:75 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds
```

Step 7: Scan All TCP Ports

Command:

```
nmap -Pn -T4 -p- 192.168.80.129
```

Explanation:

- -Pn treats the target host as online.
- -T4 increases the scanning speed.
- -p- scans all 65,535 TCP ports.

This scan helps identify services running on non-standard ports.

Observed Result:

Port	Service	Status
2222	SSH (Custom Port)	Open

```
(kali㉿kali)-[~]
└─$ nmap -Pn -T4 -p- 192.168.80.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-29 22:59 -0400
Nmap scan report for 192.168.80.129
Host is up (0.00082s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE
2222/tcp  open  EtherNetIP-1
MAC Address: 00:0C:29:96:C2:75 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 2.76 seconds
```

Step 8: Identify User Devices

The identified devices are categorized based on their roles.

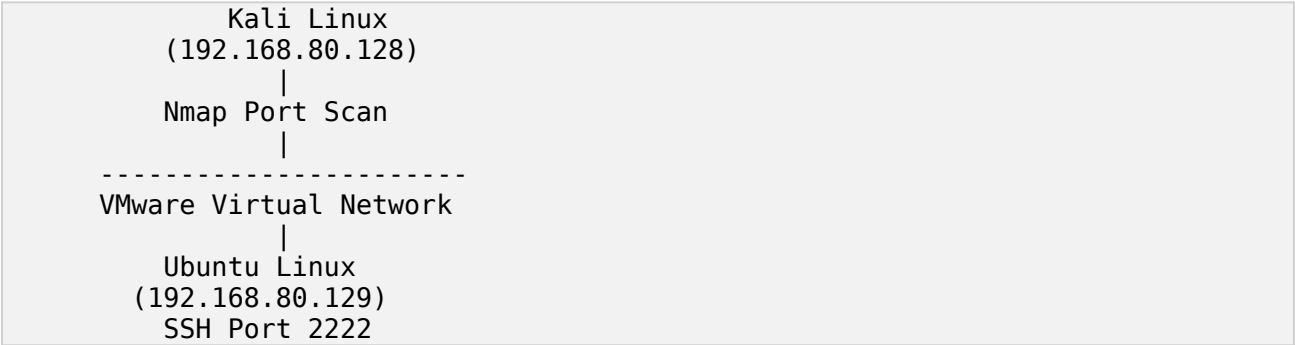
Device	Category
Ubuntu Linux	Managed Corporate Device
Kali Linux	Security Assessment Workstation

Step 9: Document Network Entry Points

The identified remote access entry points are documented below.

Entry Point	Port	Status
SSH	22	Closed
HTTPS	443	Closed
RDP	3389	Closed
SSH (Custom Port)	2222	Open

Network Diagram



Result

The remote perimeter mapping exercise was successfully completed. Nmap was installed and used to scan the target Ubuntu system. The default remote access ports (22, 443, and 3389) were found to be closed. A full TCP port scan identified an SSH service running on the custom port 2222. The remote access devices and network entry points were successfully identified and documented.

Learning Outcomes

- Understood the concept of remote perimeter mapping.
- Installed and verified the Nmap scanning tool.
- Verified network connectivity between systems.
- Performed targeted and full TCP port scans.
- Identified services running on default and non-default ports.
- Documented remote access devices and network entry points.

Conclusion

This laboratory exercise demonstrated how remote perimeter mapping is used to identify remote access services within a network. By using Nmap, the target system was scanned for common remote access ports and all TCP ports to identify services running on both default and custom ports. The exercise provided practical experience in network reconnaissance, service discovery, and documenting remote access entry points, which are essential tasks in cybersecurity assessments.